NANYANG TECHNOLOGICAL UNIVERSITY
SEMESTER 1 EXAMINATION SAMPLE PAPER
HE2003 - ECONOMETRICS I

NOTE: This paper is used to illustrate the question structure of the final exam. Therefore, the specific weightages for the topics in the course may be different from those in the upcoming final exam.

Time Allowed: 2½ hours

INSTRUCTIONS

1. This paper contains 5 questions and comprises 5 pages.
2. Answer all the questions.
3. This is a **closed-book** examination.
4. The number of marks allocated is shown at the end of each question.

THIS PAPER MUST NOT BE REMOVED FROM THE EXAM HALL

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1. Explain the following terms; Use specific examples, formulas, or graphs if necessary.

   (a) The minimization problem of the OLS estimator and its first order conditions. (5 Marks)

   (b) The (large-sample) distribution of the t-test statistic when the null hypothesis is true. (5 Marks)

   (c) Omitted variable bias. (5 Marks)

   (d) Multicollinearity and dummy variable trap. (5 Marks)

   (TOTAL: 20 Marks)

2. Consider the following linear regression model for consumption and income:

$$consumption_i = \beta_0 + \beta_1 income_i + u_i,\ u_i = (\sqrt{income_i} + 2 \cdot income_i) \cdot e_i,$$

where e_i satisfies $E[e_i] = 0$ and $Var[e_i] = \sigma^2$. In addition, e_i is independent of $income_i$.

(a) Show that this model satisfies the OLS Assumption 1. Notice that the following result holds for random variables X and Y for any function $g(X)$: $E[g(X) \cdot Y|X] = g(X) \cdot E[Y|X]$. (5 Marks)

(b) Show the relation between the R^2 from the OLS regression of $consumption_i$ on $income_i$ and the R^2 from the OLS regression of $\widetilde{consumption}_i$ on $\widetilde{income}_i$, where $\widetilde{consumption}_i = 10 \cdot consumption_i$ and $\widetilde{income}_i = 5 \cdot income_i$. (10 Marks)

(c) For testing $H_0 : \beta_1 = 0$, the t-test statistic equals 1.98. Can you determine if the value -1 is inside the 95% confidence interval of β_1? Explain. (5 Marks)

(TOTAL: 20 Marks)

3. Suppose we want to study the following linear regression model and estimate by OLS:

$$log(salary_i) = \beta_0 + \beta_1 r_{top10,i} + \beta_2 r_{11-50,i} + \beta_3 r_{51-100,i} + \beta_4 GPA_i + u_i, \qquad (1)$$

i=1, ..., n, where $salary_i$ denotes the average first-year salary of the graduates from University i. $r_{top10,i}, ..., r_{51-100,i}$ are the dummy variables for the QS rankings of universities. For example, $r_{top10,i}$ equals one if University i is among the top 10 in the QS ranking of universities and equals zero otherwise. The other dummy variables are defined similarly. GPA_i denotes the average GPA of undergraduate students at University i, and u_i denotes the error term.

(a) Explain the meaning of the OLS coefficients for β_2 and β_4. (5 Marks)

(b) Illustrate the model in (1) by using a graph. Explain why the errors in (1) may be heteroskedastic. Illustrate the heteroskedasticity in your graph. (5 Marks)

(Note: Question No.3 continues on page 3.)

(c) Suppose that the true value of β_4 in fact equals 0.03. Also suppose that the standard error of $\hat{\beta}_4$ equals 0.015. Then, for the (two-sided) t-test of $H_0 : \beta_4 = 0$ at the 5% significance level, is the Type II error of the test larger or smaller than 50%? Explain. How would you change the acceptance region of the t-test to reduce the Type II error? Explain. (5 Marks)

(d) Is it possible for the model in (1) to suffer from multicollinearity? If possible, give the specific examples of multicollinearity. If impossible, explain the reason. (5 Marks)

(TOTAL: 20 Marks)

4. Suppose that in 2021 a school district with two universities was given a sharp increase in funding. Call the two universities Eastern and Western, respectively. The district decided that the funding would go to Western, mostly to hire more teachers to reduce class sizes. To evaluate the effectiveness of the increased funding, the district collected random samples of the students in 2020 and 2022 at both universities. The measure of student performance is a standardized test score given to all undergraduate students in the district.

   (a) Let $score$ be the test score. Without controlling for other factors, write down a difference-in-differences model using linear regression with an interaction term that allows you to determine whether the additional funding led to improved test scores. Define precisely all the regressors in your model. Explain the specific meaning of all the parameters in the model. (10 Marks)

   (b) What is the key assumption of the difference-in-differences model? Explain. (5 Marks)

   (c) How might estimating the model in (a) be biased if Western gets an influx of better students because of the increased funding? Explain. (5 Marks)

(TOTAL: 20 Marks)

(Note: Question No.5 continues on page 4.)

5. Consider an equation to explain performance bonuses (bonus) of senior managers in terms of annual firm sales (sales), return on equity (roe, in percentage form), and return on the firm's stock (ros, in percentage form):

$$log(bonus) = \beta_0 + \beta_1 log(sales) + \beta_2 roe + \beta_3 ros + u. \tag{2}$$

   (a) Using the data, the following equation was obtained by OLS:

$$\begin{aligned} \widehat{log(bonus)} \quad = \quad & 4.3 + 0.3 log(sales) + 0.018 roe + 0.00025 ros, \quad n = 221, \quad R^2 = 0.3. \\ & (0.3) \;\; (0.35) \qquad\quad (0.0036) \qquad (0.0005) \end{aligned} \tag{3}$$

   By what percentage is $bonus$ predicted to increase if ros increases by 50 units? Furthermore, explain the meaning of the coefficient for $log(sales)$.
   (5 Marks)

   (b) Suppose we want to test the hypothesis that none of the explanatory variables in (3) has an effect on the performance bonus. State the null hypothesis and (two-sided) alternative hypothesis. Explain what test statistic you would use here and how to find its corresponding critical value. State the critical value at the 5% significance level by using the critical value tables on page 5. Compute the value of your test statistic and state whether the null hypothesis can be rejected at the 5% significance level.
   (5 Marks)

   (c) Suppose we want to allow for the difference in the effect of roe on $log(bonus)$ between IT firms and Non-IT firms. How can we extend the model in (2) to allow for different slopes of roe and different intercepts for the two types of firms? Write down your new model and explain the meaning of all the new regressors and parameters added by you. (10 Marks)

(TOTAL: 20 Marks)

Table 1. Critical values for the χ^2 distribution

	Significance Level		
Degrees of Freedom	**10%**	**5%**	**1%**
1	2.71	3.84	6.63
2	4.61	5.99	9.21
3	6.25	7.81	11.34
4	7.78	9.49	13.28
5	9.24	11.07	15.09
6	10.64	12.59	16.81
7	12.02	14.07	18.48
8	13.36	15.51	20.09
9	14.68	16.92	21.67
10	15.99	18.31	23.21
11	17.28	19.68	24.72
12	18.55	21.03	26.22
13	19.81	22.36	27.69
14	21.06	23.68	29.14
15	22.31	25.00	30.58

Table 2. Critical values for the $F_{m,\infty}$ distribution

	Significance Level		
Degrees of Freedom	**10%**	**5%**	**1%**
1	2.71	3.84	6.63
2	2.30	3.00	4.61
3	2.08	2.60	3.78
4	1.94	2.37	3.32
5	1.85	2.21	3.02
6	1.77	2.10	2.80
7	1.72	2.01	2.64
8	1.67	1.94	2.51
9	1.63	1.88	2.41
10	1.60	1.83	2.32
11	1.57	1.79	2.25
12	1.55	1.75	2.18
13	1.52	1.72	2.13
14	1.50	1.69	2.08
15	1.49	1.67	2.04

END OF PAPER